Native Binary Rings, Observation Depth, And the Memory of Continuation

Direct Addressing, Phase Synchronization, and Exact Alias Resolution in a Partner-Ordered 64-Bit Ring

Driven by Dean A. Kulik

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Abstract

A binary word ring with a specified native partner ordering contains 2^n word addresses and $n2^n$ serialized bit-slice positions. Although its complete waveform is exponentially large in word width n, address recovery can be performed by an inverse grammar that evaluates only candidate word-boundary alignments. This study develops the formal reader contract, derives its reversible native successor and reflection laws, and measures synchronization and memory across a sequence of exact computational experiments culminating in the n = 64 construction. A known 128-address grid on R_{64} has 8,192 combined word/phase origins. On that qualified domain, 127 consecutive bits identify all origins; with the same sources but competitors admitted from the full 2^{70}-slice ring, 256 consecutive bits are necessary and sufficient. At 127 bits, 5,743 of the 8,192 sources are uniquely identified without the grid qualifier, with 2,451 off-grid alias pairs affecting 2,449 sources. The longest alias shares 255 bits and separates at bit 256. An independent sparse reader needs 55 widely separated samples to synchronize unknown phase and grid index; both readout modes retain the result as a 13-bit coordinate. Additional within-word and seam-crossing readers show 62- versus 14-bit one-shot acquisition for 128 known-phase sources, while the retained address requires seven bits. Exact counterexamples prevent extrapolation: R_6 has a 24-bit alias and requires 25 bits, disproving the unqualified bound $w^*(n) \le 4n$. The paper distinguishes algebraic consequences, finite-domain execution results, corrected errors, and unproved all-width or physical interpretations. The central operational result is that synchronization depth is a property of the reader and its candidate domain, whereas memory is the minimal future-relevant address once location has been established.

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

Keywords: native binary ring; phase synchronization; continuation equivalence; inverse address; partner reflection; finite-state reader; information retention; alias length; computational geometry.

1. Introduction

Computational descriptions often conflate three different entities: the full domain of possible configurations, the route by which an observation reaches one configuration, and the information that must be kept after that observation. In the NEXUS continuation-ledger program [1, 2], a transformation has its own inverse or retained way home, and a readout is explicitly typed by its aperture, sampling route, and admissible comparison domain. Those distinctions motivate a precise study of how a reader identifies its position in a known but exponentially large structured binary ring.

The question investigated here is not whether arbitrary data can be reconstructed from a few bits, nor whether a physical signal violates Nyquist–Shannon sampling theory. It is narrower and exact: when the complete finite generator R_n is shared, how many observations are needed to distinguish a starting address? How does this depth change when the reader is phase aligned, unaligned, confined to a grid, or tested against all possible native addresses? Which extra bits reflect actual algebraic boundary dependencies, and which represent knowledge of the reader contract?

The native ring considered here is not the conventional binary count order and not the complemented shift register τ_n used elsewhere in the NEXUS corpus. Confusing these motions led to earlier erroneous extrapolations. The present paper explicitly defines their separate types and treats prior numerical results as propositions only within their actual execution domains.

The contributions are: (i) formal reconstruction of the partner-ordered ring and its rank/inverse; (ii) a direct-alignment inverse algorithm whose candidate count is bounded by n rather than 2^n; (iii) exact gcd-based classification of uniform phase sampling; (iv) separate acquisition, retention, and build-cost ledgers for readers; (v) an exhaustive 8,192-source R_{64} competitor audit and explicit 255-bit alias; (vi) the representative-count derivation of the 3/8 seam; and (vii) a width-6 counterexample limiting any universal $4n$ claim.

2. Formal domain: words, partners, and the native circumference

2.1 Words and partner involution

For integer $n \geq 2$ define the binary word domain $X_n = \{0,1\}^n$ and $N_n = |X_n| = 2^n$. For $x = (x_0,\ldots,x_{n-1})$, $J_n(x)$ reverses the order of bits and $C_n(x)$ complements each bit. Define the native partner P_n by reversing nonpalindromes and complementing palindromes:

$$P_n(x) = J_n(x) \text{ if } J_n(x) \neq x; \quad P_n(x) = C_n(x) \text{ otherwise.} \qquad (1)$$

The map P_n is a fixed-point-free involution. For a nonpalindrome it exchanges x and $J_n(x)$; for a palindrome it exchanges x with the distinct palindrome $C_n(x)$. Thus X_n splits into 2^{n-1} disjoint pairs. Let $c(x) = \min\{x, P_n(x)\}$ under the ordinary lexicographic order of n-bit words, and let $r_0 < r_1 < \ldots < r_{m-1}$ be all canonical representatives, with $m = 2^{n-1}$. This order is only a reproducible ranking convention, not the claim that the native ring successor increments the represented binary value.

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2.2 Native ordering and its seam

$$R_n[a] = r_a \ (0 \le a < m-1); \qquad R_n[m-1] = P_n(r_{m-1}); \qquad (2)$$

$$R_n[m] = r_{m-1}; \qquad R_n[a] = P_n(r_(N_n-1-a)) \ (m < a < N_n). \qquad (3)$$

Equation (3) uses the canonical representative indexed by N_n-1-a; the partner relation can also be written directly in address form:

$$R_n[N_n - 1 - a] = P_n(R_n[a]) \quad \text{for every a.} \qquad (4)$$

The two central addresses exchange the expected representative/partner ordering. This is the native equatorial seam of the retained code. Equation (4) is the operational partner law; the forward native successor is the circular increment of the R_n address, not bitwise binary increment on the word.

For n = 2, the supplied base ring is R_2 = 00 | 10 | 01 | 11, with serialized waveform B_2 = 00100111. For n = 3, R_3 = 000 | 001 | 010 | 110 | 011 | 101 | 100 | 111. These examples are small enough to inspect directly and correspond to the established native construction [2].

2.3 Circumference and bit-slice reader

$$S_n = nN_n = n2^n; \quad t = na + p, \quad 0 \le p < n. \qquad (5)$$

Define the circular binary waveform by $B_n(t)$ = the pth bit (from the most-significant side) of $R_n[a]$, where a = ⌊t/n⌋ and p = t mod n, with t reduced modulo S_n. Thus R_{64} contains $N_{64} = 2^{64}$ words but $S_{64} = 2^{70}$ bit-slice positions. Conflating n, N_n and S_n changes the sampling domain by exponential factors.

2.4 Distinct clocks

The NEXUS twisted clock $\tau_n(x_0 \ldots x_{n-1}) = (x_1 \ldots x_{n-1}, \neg x_0)$ has $\tau_n{}^n = C_n$ and $\tau_n{}^{2n} = I$ [1]. Its order divides 2n. The native address successor T_n is instead an N_n-cycle on words. Spectral anti-periodicity under τ_n does not therefore transfer automatically to the serialized native waveform B_n. These are different transformations even though both operate on the same binary word domain.

3. Exact compiler: native rank, inverse, and address continuation

3.1 Canonical ranking and unranking

Let $L_n(x)$ be the native word address of x, and $U_n(a) = R_n[a]$ its inverse. Canonical representative ranking counts how many representatives precede c(x), using paired inward bit comparisons. The dynamic-programming state retains the current comparison outcome between a prefix and its reversed counterpart and, on palindrome branches, the leading face bit. The supplied standard-library implementation tests at most two bit choices per unassigned position in each inward pair. The rank procedure applies prefix counts successively to the bits of x. Unlike enumerating all 2^n words, it computes a word address with polynomially many finite integer operations in n.

3.2 Word successor and exact way home

$$T_n = U_n \circ (+1 \bmod 2^n) \circ L_n, \quad T_n{}^{-1} = U_n \circ (-1 \bmod 2^n) \circ L_n. \qquad (6)$$

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The "carry chain" lies in the rank/address coordinate. If address bits are numbered from least significant to most significant, setting $c_0=1$ and $c_{j+1}=c_j \wedge a_j$, $a'_j=a_j \oplus c_j$ implements the increment. Applying U_n transforms the incremented address back to the native word. Because L_n and U_n are bijections, T_n is a bijection and $T_n^{Nn} = I$. The operation is not in general an affine update of the n word bits.

Exact finite Boolean influence matrices for T_n have one zero at $n = 2$ and $n = 3$, and all entries one for $n = 4,\ldots,12$ in the recorded audits [2]. These are existential dependency matrices: a 1 means some input mutation can affect an output position. They are not linear matrices implementing T_n, and the all-width saturation statement is not established by the finite check.

3.3 Inverting an unaligned observation

For a W-bit observed string s with $W \geq 2n-1$, every starting phase guarantees the existence of at least one complete n-bit word inside the observation. A reader can try the n possible word-boundary offsets $\delta = 0,\ldots,n-1$. For each δ, it extracts an n-bit candidate w, calculates $a=L_n(w)$, proposes the absolute bit start $t=na-\delta \pmod{S_n}$, and checks whether the native generator reproduces all W observed bits. Returned candidates are all valid native origins, by construction.

Algorithm 1. Generic native inverse reader

```
INPUT: observed bits s, width n, shared native rank Ln and unrank Un
REQUIRE: length(s) W ≥ 2n − 1
CANDIDATES := ∅
FOR δ = 0,…,n−1:
    w := binary word s[δ : δ+n]
    a := Ln(w);  t := (n*a − δ) mod (n*2^n)
    IF native_read(t,W) = s: CANDIDATES := CANDIDATES ∪ {t}
RETURN CANDIDATES
```

The algorithm has at most n phase hypotheses, not 2^n word hypotheses. With polynomial-time native rank/unrank and a window of polynomial length, its arithmetic work is polynomial in n and W. The reader may still return more than one candidate: computational tractability and information-theoretic identifiability are distinct conditions. If $W < 2n-1$, the guaranteed-complete-word argument no longer applies to every possible phase.

4. Reader contracts and synchronization mathematics

4.1 Source domain versus competitor domain

For any finite set G of source bit starts and any comparison set H containing G, define the length-W observation $A_W(t) = (B_n(t),\ldots,B_n(t+W-1))$. A source $t \in G$ is uniquely located relative to H when $A_W(t)=A_W(u)$, $u \in H$, implies $u=t$. Define the contract threshold as the least W that uniquely locates all sources. If this property never holds, the threshold is infinite. The full-ring quantity $w^*(n)$ is obtained when $G=H=\mathbb{Z}/S_n\mathbb{Z}$.

$$d(G,H) = \min\{W : \forall t \in G, \forall u \in H, A_W(t)=A_W(u) \Rightarrow t=u\}. \quad (7)$$

Increasing H can only increase or preserve d(G,H); restricting H can only decrease or preserve it. Therefore the depth inferred from a prequalified 128-address grid is not automatically the depth required against the entire native ring.

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4.2 Longest-common-prefix theorem

Let LCP(t,u) denote the number of initial bit slices shared by the infinite periodic readings beginning at t and u, up to the first differing bit. If all competitor pairs eventually separate, the exact threshold is:

$$d(G,H) = 1 + \max\{LCP(t,u) : t \in G, u \in H, u \neq t\}. \qquad (8)$$

Proof. A pair sharing L initial bits cannot be distinguished by a W-bit observation for $W \leq L$; at $W = L+1$ its first differing bit is observed. Taking the maximum over admissible pairs gives necessity and sufficiency. The formula requires eventual distinguishability; otherwise the maximum may be infinite. This is the criterion used to turn alias data into sharp finite reader depths.

4.3 Uniform strides and phase locking

For aperture-one sampling on the full bit circumference with starting slice t_0 and stride s, define $y_j = B_n(t_0 + js)$. Elementary cyclic-group arithmetic gives two orbit lengths:

$$P_slice = S_n / \gcd(S_n, s); \quad P_phase = n / \gcd(n, s). \qquad (9)$$

The first is the number of distinct bit addresses visited before the route closes; the second is the number of intra-word phases visited. For $n = 64$, a 128-samples-per-revolution route has $s = S_{64}/128 = 2^{63}$, giving $P_phase = 1$ and exactly 128 visited slice addresses. All samples stay in a single bit phase even while the word address changes by 2^{57} between samples.

Nyquist–Shannon reconstruction assumes a specified signal class, sampling timebase, and bandlimit [4,5]. A 64-cycle sinusoid sampled 128 times per full revolution sits at the Nyquist boundary; its quadrature is not generally identified by those samples alone. This conventional statement does not imply that 128-bit word width means 128 samples or that the native bit waveform is bandlimited. Ring-native recovery can exploit the known discrete generator, but that is an additional model constraint, not a violation of sampling theory.

4.4 Polyphase and reflected readout

The bitstream decomposes into n word-phase lanes $L_p(a)$, one for each $p = 0,\ldots,n-1$, where $L_p(a)$ is bit p of $R_n[a]$. Interleaving all n lanes in native word order gives B_n exactly. The partner law provides a second valid address relation:

$$B_n(S_n - 1 - t) = B_n(t) \oplus \chi_pal(R_n[\lfloor t/n \rfloor]). \qquad (10)$$

Here χ_pal is one precisely for palindromic words. Complement occurs because P_n uses C_n on palindromes and J_n otherwise. This allows the reflected address of any sampled bit to be calculated directly, and sometimes its bit value when the palindrome flag is known. The flag itself may require the native grammar or additional information; reflection does not reconstruct arbitrary unseen physical data merely from a single observed bit.

Across 128 phase-5 samples on R_{64}, the reflected route lies at phase 58 with opposite stride direction. The audit yielded 127 equal bit values and one complemented value, corresponding to the palindromic origin. This checks the exact partner relation on the declared sampling route [2].

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5. Memory as a continuation-equivalence class

5.1 What must be retained

Let Q be a specified readout and let C be the set of permitted future continuations. For two past observation histories h and h′, define h~Qh′ if Q(hc)=Q(h′c) for every c∈C. This is the established continuation-equivalence construction used for finite-state recognition [3,6] and already adopted in the NEXUS §77 ledger [1]. The memory state can be represented by the resulting equivalence class [h]~Q. Importantly, this is relative to the declared future queries. A winding count is unnecessary for predicting a periodic output phase but necessary if the question explicitly asks how many complete circuits have occurred.

$$M_Q(h) = [h]\sim Q, \quad h\sim Qh' \Leftrightarrow \forall c\in C,\ Q(hc)=Q(h'c). \qquad (11)$$

A finite set of K distinguishable terminal origins requires at least $\lceil \log_2 K \rceil$ bits in a fixed-width address code. This is a storage lower bound after the origin is known. It does not bound the number of noisy, constrained, or poorly aligned measurements needed to acquire that origin, nor the computational cost of building the decoder. Distinguishable reading prefixes can be much longer than the address eventually retained.

5.2 Recipe, operator, and noun

The independent §105–106 route experiments distinguished an ordered opcode recipe from its composed transformation and its particular payload [1,2]. For example, the fixed three-bit reversible verb sequence C01→SW12→X2→SW02 has T(a,b,c)=(a⊕b⊕1,c,a), with inverse (u,v,w)↦(w,u⊕w⊕1,v). It acts on all eight three-bit nouns without changing the recipe; its permutation has a seven-cycle and one fixed point. Adding an independent toggling sheet produces a different reversible four-bit extension with 14- and two-cycles. Such route grammar illustrates the separation of execution history from readout, but it is not the native R_n successor or a derivation of physics.

The paper therefore treats "memory is a recipe" as a precise, typed statement: the route grammar can be shared, the acquired address can be retained compactly, and full historical reconstruction requires whatever additional distinctions the query demands. The minimal reader state is neither the full noun inventory nor automatically a lossless code for an arbitrary input.

6. Experimental design and reproducibility

6.1 Source programs and measurements

The experiments derive from the original standalone NEXUS §§170–179 Python programs and their printed execution logs [2]. They use the partner function, dynamic-programming canonical rank, native unrank, and direct bit-window reader; no exhaustive traversal of R_{64} is performed. Small rings are enumerated exhaustively to certify exact minima. On R_{64} the finite source set is explicitly fixed at 128 word addresses and 64 phases, for 8,192 total tested sources. Where global competitors are admitted, the inverse algorithm generates candidate word-boundary offsets using the rank grammar rather than enumerating 2^{70} full-ring bit starts.

$$G_{64} = \{(a_j,p): a_j=j\cdot 2^{57},\ 0\le j<128,\ 0\le p<64\}. \qquad (12)$$

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Each (a_j,p) has absolute bit-slice address $t=64a_j+p$. The sparse temporal route advances by 2^{63} bit slices per observation, so the word hop is 2^{57}. The contiguous route advances by one bit slice per observation. These are not the same physical or computational measurement contract.

6.2 Exact comparisons and finite certification

For small n, the verifier materializes the complete native circumference and compares all cyclic W-bit prefixes. On R_{64}, the generic inverse examines 64 possible word-boundary alignments per source. Across 8,192 sources this gives 524,288 nominal alignments; caching is legitimate because the trial n-bit word depends only on $p+\delta$. Exactly 16,159 distinct 64-bit trial subwords occurred in the source set, allowing exact native rank evaluation and regeneration of every surviving alias. Caching changes execution time, not the candidate domain or equality test.

At each window length W, uniqueness is computed from actual prefix equality, not inferred from the scarcity of matches. Alternative addresses are decoded into their native word and bit phase and independently passed through the forward `read` routine for a way-home check. This is a finite-domain exhaustive audit of 8,192 specified source origins relative to the entire ring as a competitor domain; it is not an all-source census of 2^{70} starts.

6.3 Reporting conventions

Statements marked STANDS are algebraic theorems or exact consequences of definitions; RUNS reports finite verified executions with their domain stated; DEAD END marks refuted proposed routes or hypotheses; OPEN identifies a theorem or cross-domain bridge that has not been supplied. The §178 initial SHA-256 prediction was sealed before execution but contained a wrong exponent in the MSB-count expression, subsequently corrected. The existence of that seal does not turn the corrected assertion into a prediction made before reading the data. The §179 exception audit is retrospective and is reported as such. No claim of independent physical validation, SHA inversion, or P-versus-NP resolution is made.

7. Baseline: full-ring synchronization through small and medium widths

Define $w^*(n)=d(\mathbb{Z}/S_n\mathbb{Z},\mathbb{Z}/S_n\mathbb{Z})$. Exact exhaustive checks give the following sequence for n=2,…,17; the n=17 value is supplied by the independent held-out depth check, while §§170 and 179 provide the small-domain inverse/read validations [2].

Table 1. Complete native-ring synchronization depths measured at widths 2–17.

n	Word count 2^n	Bit circumference $n2^n$	Exact $w^*(n)$
2	4	8	5
3	8	24	11
4	16	64	13
5	32	160	16
6	64	384	25
7	128	896	27
8	256	2,048	32

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n	Word count 2^n	Bit circumference $n2^n$	Exact w*(n)
9	512	4,608	36
10	1,024	10,240	40
11	2,048	22,528	44
12	4,096	49,152	48
13	8,192	106,496	52
14	16,384	229,376	56
15	32,768	491,520	60
16	65,536	1,048,576	64
17	131,072	2,228,224	68

At n=8 through n=17, the measured values equal 4n. However, n=6 requires 25=4n+1, so "w*(n) ≤ 4n for every n" is false. The finite observations do not prove w*(n)=4n for every n≥8. Establishing a uniform upper bound from the canonical representative and seam structure is an OPEN mathematical problem in this program.

7.1 Exact width-6 alias and refutation of a universal 4n ceiling

Two different bit starts on the same native R_6 circumference

t = 84 (word 14, phase 0)	001111010001010010010011	**0**
u = 314 (word 52, phase 2)	001111010001010010010011	**1**
	← 24 identical bits →	bit 25

Figure 1. Certified width-6 counterexample. Bit starts 84 and 314 have the same first 24 bits and different 25th bits; the exhaustive 384-origin inverse gives w(6)=25.*

At width six, positions t=84=(14,phase 0) and u=314=(52,phase 2) both emit 001111010001010010010011 over 24 consecutive bit slices. The next bit is 0 from the first origin and 1 from the second. Exhaustive tests of all 384 starts find no duplicate 25-bit readings; therefore w*(6)=25. This proves the negative claim without extrapolation. The two sources belong to different native word sequences, and their agreement extends across distinct word-boundary alignments. Their ownership is thus a phase alias, not a missing binary value.

8. Width-64 sampling field: the known 128-address grid

8.1 Seven-bit route index and sampled words

The 128 grid addresses are $a_j=j2^{57}$. Under the shared grammar U_{64}, each sampled word $R_{64}[a_j]$ is distinct. Therefore its source index j is a seven-bit coordinate once the route is known. Conversely, $L_{64}(R_{64}[a_j])=a_j$ and division by 2^{57} returns j exactly. This 7-bit address does not contain the 64-bit word as independently transferable information; the word is regenerated by the shared grammar. If the rank/unrank grammar is unknown or changes, the same address may no longer be sufficient.

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8.2 One-bit temporal readers

For each phase p, define the 128-element cyclic binary lane $v_p(j)=bit_p(R_{64}[j2^{57}])$. Each lane has primitive period 128 in the computed grid. Its minimum cyclic prefix depth is the shortest length that uniquely distinguishes its 128 starting indices. The 64 measured depths range from 11 to 52 samples, although each lane ultimately retains only seven bits of j. Phase 13 is among the fastest (11 samples), while the originally chosen phase 5 requires 15, and phase 0 requires 52. The waveform is not a de Bruijn sequence of order seven: uniqueness at seven samples would be required for that stronger property, which was not observed.

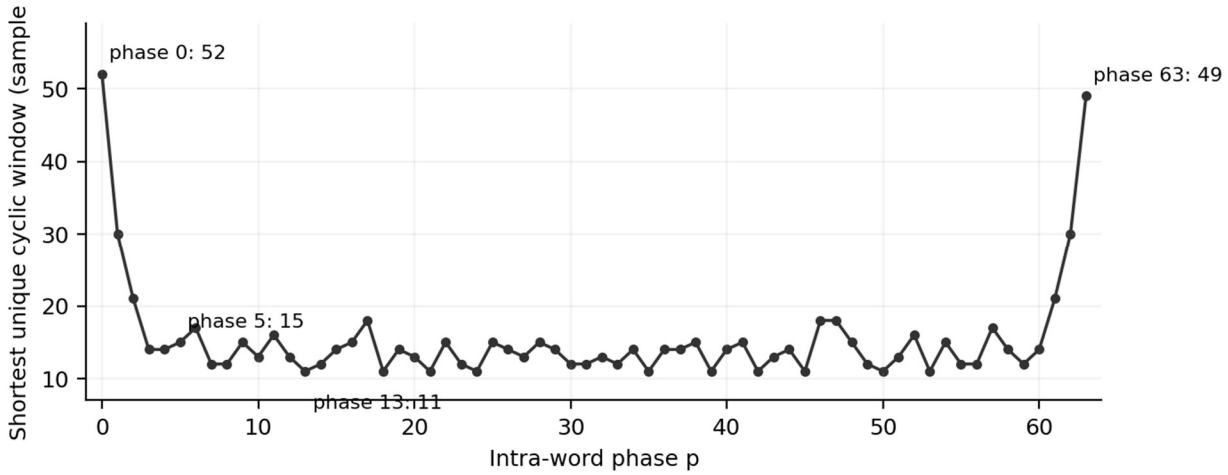


Figure 2. Exact minimal synchronization depth for the 64 phase lanes on the same 128-address grid. Lane depth varies from 11 to 52 temporal samples while the route memory remains seven bits.

For phase 5, at 14 samples there are 126 different prefix strings but only 124 uniquely identified starts; two pairs still collide. All 128 starts become unique at 15. The distinction between number of distinct prefixes and number of uniquely identified origins was corrected in §175 and must be maintained for interpretation.

8.3 Seven-bit memory after acquisition

The same seven-bit j is returned by the phase-13 and phase-5 readers. In the latter case, 15 widely separated one-bit observations are sufficient in the worst case to acquire the index, but the acquisition buffer need not be retained once j is bound and the future query uses the same shared route. Because sparse samples are separated by 2^{63} bit slices, their acquisition depth cannot be compared to contiguous bit-window depth as if they were the same physical exposure time or traverse length.

9. Single-snapshot readers: word interior versus seam

9.1 Forced bit positions within a word

For the 128 grid words, five source pairs have Hamming distance one. A reader using a fixed set of bit positions must include the unique distinguishing phase for each such pair. The forced positions are 2, 61 and 62. Consequently, any nonwrapping contiguous window entirely within a word that contains all forced positions must have length at least 62, spanning phases 1 through 62 when the final unresolved pair is included. The existing exact experiment checks that phases [1,62] distinguish all 128 sampled words; a 61-bit interval fails. This is a lower bound for the specified contiguous-within-word reader, not for arbitrary, noncontiguous detectors.

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9.2 Fourteen-bit seam reader

$$Q_{14}(a) = \mathrm{suffix}_2(R_{64}[a]) \,\|\, \mathrm{prefix}_{12}(R_{64}[a+1]). \qquad (13)$$

Starting at phase 62, a 14-bit aperture reads two bits from the current word and twelve bits from the next native word. All 128 source addresses yield distinct Q_{14} signatures; 13 bits are insufficient because indices 72 and 74 share 0111000011101 and differ on the 14th bit (1 versus 0). A prefix reader based on these 128 signatures has 293 reported reachable nodes and 128 terminal owners; mean acquisition depth under a uniform source distribution is 1014/128=7.921875 bits, with maximum 14. The shared grammar returns the seven-bit j at the terminal state.

The 62-versus-14 result therefore compares different fixed apertures: inside-word versus boundary-crossing. It shows that the native successor contributes additional discriminating structure. It does not mean that fourteen arbitrary bit samples suffice to reconstruct arbitrary 64-bit data in the absence of the declared native grammar and finite source family.

9.3 Forced differences as the shape of retained context

The Hamming-one pairs identify which source distinctions are invisible unless particular phases are observed. The seam reader does not recover their missing bits by convention. It changes the observation contract to include the native successor and lets that additional lawful continuation separate the remaining classes. In continuation-equivalence language, the memory class becomes singleton on the 128-source domain when the 14-bit prefix is read. The resulting address is a sufficient future-state summary; the 14 observation bits are transient acquisition history.

10. Unknown phase: the 8,192-origin grid experiment

10.1 Thirteen-bit coordinate

If both the grid index j and intra-word phase p are unknown, but their domains remain $j \in \{0,\ldots,127\}$ and $p \in \{0,\ldots,63\}$, the origin set has 128×64=8,192 elements. The fixed-length binary coordinate (j,p) uses exactly 7+6=13 bits. Provided the native ring and grid are shared, it returns the bit-slice address $t=64(j2^{57})+p$. This is the retained location memory, not an assertion that thirteen sampled bits always determine it.

10.2 Sparse unknown-phase reader

The stride-2^{63} reader is run for every one of the 8,192 (p,j) origins. All complete 128-sample streams are distinct. The largest shared prefix among unequal streams has length 54, so 55 samples suffice in the worst case; the mean adaptive prefix depth is 15.4169921875 samples. The last collision is (p=0,j=0) against (p=63,j=80), which agree for $0^{49}1^5$ and differ on observation 55. The final two digits are 1 and 0, respectively. All 8,192 origin returns passed the compiled decoder [2].

10.3 Contiguous unknown-phase reader

With the same source and competitor grid, but a contiguous one-bit-at-a-time aperture, all 8,192 starts are identified by W=127 bit slices, and two still collide at 126. The final pair is (p=0,j=0) versus (p=1,j=0). This result follows directly from the origin sequence $R_{64}[0]=0^{64}$, $R_{64}[1]=0^{63}1$: the two one-bit-shifted readings share their first 126 zeros and separate on bit 127. Mean adaptive depth over these 8,192 sources is 18.0020751953125 bits. The decoder again retains 13 bits after unique binding.

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These 55 and 127 thresholds are exact only for the declared G_{64}-to-G_{64} contract. They do not say that every full-ring location outside the grid is excluded by such prefixes. The following unfiltered audit measures precisely that difference.

11. Exact 3/8 boundary and the 54-sample sparse collision

11.1 Derivation from native canonical representatives

Consider canonical representatives whose most-significant bit (MSB) is zero for even n≥4. Words beginning in 0 and ending in 1 are automatically lexicographically smaller than their reversals; there are 2^{n-2} of these. Among words beginning and ending in 0, reversal forms two-element orbits except for palindromes. The count of those latter words is 2^{n-2}, while the number of palindromes with both endpoints fixed to zero is $2^{n/2-1}$. The reversal orbits contribute $(2^{n-2}-2^{n/2-1})/2$ canonical representatives; the palindromes contribute one canonical member of each complement-pair, all with leading zero. Summing gives the exact threshold:

$$C_0(n) = 3\cdot 2^{n-3} + 2^{n/2-2} \qquad (n \text{ even}, n \geq 4). \qquad (14)$$

$$C_0(64) = 3\cdot 2^{61} + 2^{30} = 6{,}917{,}529{,}028{,}714{,}823{,}680. \quad (15)$$

The count is a number of canonical representatives before the MSB=1 class in the ascending representative half. Relative to all 2^{64} word locations, $C_0(64)/2^{64} = 3/8 + 2^{-34}$. When sampled in steps of 2^{57} native word addresses, the first grid index beyond the boundary is:

$$j_1 = \lceil C_0(64)/2^{57} \rceil = \lceil 48 + 2^{-27} \rceil = 49. \qquad (16)$$

Therefore the phase-zero (MSB) grid waveform begins with exactly 49 zeros, followed by fifteen ones up to the left/right boundary of the 128-address grid. The right-hemisphere LSB values are related to left-hemisphere MSBs by the partner reflection law, with the essential off-by-one in reflected word addresses: the facing address of $k\cdot 2^{57}$ is $(128-k)\cdot 2^{57}-1$, not $(128-k)\cdot 2^{57}$. Reflecting indices 80,…,127 and passing through index zero yields the 49-zero onset of the phase-63 waveform, with the next five observations matching phase zero. The 55th separates the two origins, as measured.

11.2 Correction retained in the record

The originally sealed §178 prediction misstated the count as $3\cdot 2^{62}+2^{31}$, and a first attempted correction still used the incorrect palindrome exponent. The executed code and final ledger corrected the expression to $3\cdot 2^{61}+2^{30}$. The observed boundary $j_1=49$ and sparse collision remained unchanged. This paper explicitly derives (14) rather than relying on the earlier mislabeled numerical expression. The correction is substantive and must accompany any citation of the pre-execution seal.

12. Unfiltered competitors on the full R_{64} ring

12.1 Why the competitor domain matters

Keep the 8,192 source origins G_{64} unchanged, but expand the admissible competitor set to every one of the 2^{70} serialized positions. Generic inverse reading produces at most 64 candidate word-boundary alignments for each W-bit source observation. A candidate can be an actual alternative origin outside the source grid. Hence the unique-source count need not equal the 8,192 grid-qualified count at W=127.

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12.2 Exhaustive alias census

The §178 audit evaluated all 8,192 sources against every candidate produced by the native 64-phase inverse grammar. At W=127, 5,743 source origins are uniquely identified; the other 2,449 have at least one off-grid competitor. There are 2,451 alias pairs in total, because 2 sources have two different alternatives. After enlarging the aperture, ambiguity decreases monotonically, and all 8,192 sources become unambiguously identified at W=256. The sharp bound for this contract is proved by the 255-bit witness given below, together with the exhaustive 256-bit absence of aliases.

Table 2. Unfiltered full-ring competitor results for 8,192 selected R_{64} sources.

W (bits)	Unique out of 8,192	Ambiguous	Notes
127	5,743	2,449	2,451 aliases; 2 triple cases
128	5,806	2,386	63 further origins resolved
134	6,061	2,131	
192	7,977	215	3n
200	8,029	163	
250	8,179	13	
252	8,185	7	
253	8,188	4	
254	8,190	2	
255	8,191	1	One final alias
256	8,192	0	4n; exact selected-source threshold

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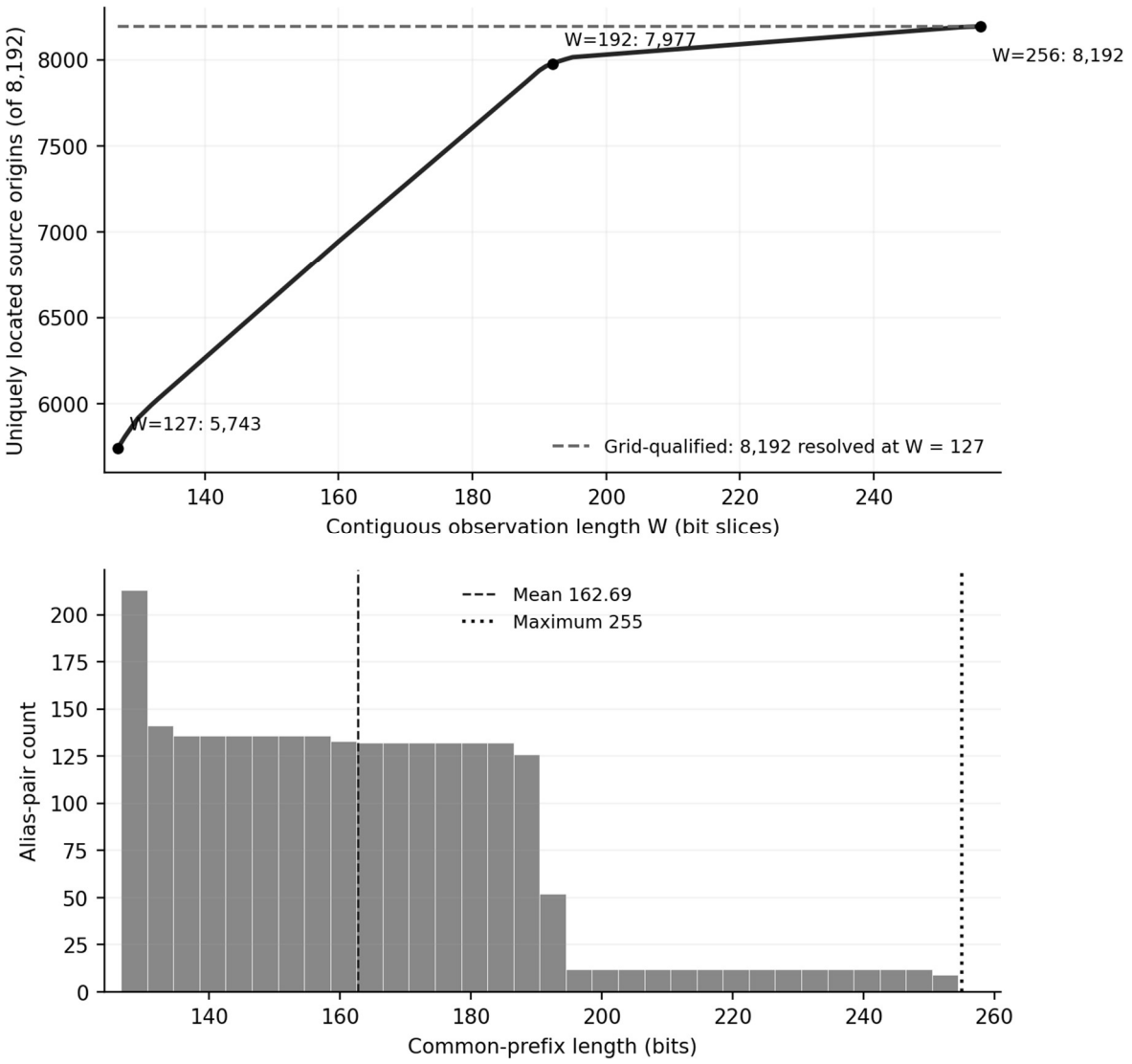


Figure 3. Exhaustive §178 source-origin audit. Top: grid-qualified (dashed) and unrestricted-competitor (solid) unique counts for the same 8,192 sources. Bottom: common-prefix lengths for 2,451 off-grid alias pairs. Competing locations are drawn from the full R_{64} circumference; source origins remain grid-selected.

The 2,451 prefix lengths have mean 162.6863, median 160, minimum 127, maximum 255, and reported sample standard deviation 25.8573 bits. These are statistics of the alias pairs surviving at 127 bits, not a probability distribution of arbitrary ring addresses. They must not be represented as a property of every location on R_{64}.

12.3 Exact 255-bit champion alias

The final off-grid competitor belongs to grid source (j=87,p=2), with source word address a=12,538,021,362,599,460,864 and bit-slice address t=802,433,367,206,365,495,298. Its alternative has word address b=6,138,513,333,050,609,951, phase 63, and bit-slice address u=392,864,853,315,239,036,927. Their first 255 consecutive bits are identical; at bit 256 the true route emits 0 and the alias emits 1. This exact one-bit separation proves W≥256 is necessary for the selected-source versus full-competitor contract.

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Inspection of four adjacent native words reveals the mechanism. The true right-hemisphere route moves mainly in its leading bits while holding a 62-bit suffix; the alternative left-hemisphere route moves mainly in trailing bits while holding a 61-bit prefix. The offset alignment causes these two long fixed bands to overlap across successive word seams. The alias is a deterministic consequence of which information the reader has not yet crossed, rather than an unassigned byte.

12.4 Two triple-candidate origins

At W=127, exactly two grid source origins have three possible native locations, counting the true location. The origin (j=41,p=63) has alternatives at (word address 8,394,709,705,847,262,045; phase 0), sharing 127 bits, and (10,128,734,440,462,975,044; phase 4), sharing 190 bits. The origin (j=57,p=0) has alternatives (5,188,146,768,617,014,302; phase 63) and (12,612,475,992,741,232,481; phase 3), each sharing 127 bits. Regeneration from the native ring returns every source and alternative and verifies the next-bit separation. After W=128 the (57,0) source has no remaining alias; after W=191 the (41,63) source is also unique.

Table 3. All four additional candidate locations belonging to the two triple-owner sources.

True grid origin	Alternative word address	Alternative phase	Shared prefix
(41,63)	8394709705847262045	0	127 bits
(41,63)	10128734440462975044	4	190 bits
(57,0)	5188146768617014302	63	127 bits
(57,0)	12612475992741232481	3	127 bits

This multiple-candidate observation does not imply that the original system contains multiple simultaneous physical realities. It means the specified W-bit readout, when compared to all permitted starts, admits more than one consistent native address until a later bit separates the candidates.

13. Comparative economics of acquisition, retention, and compilation

Table 4 consolidates the reader regimes and avoids comparing non-equivalent costs as if they were interchangeable. "Acquisition" is measured in observations or consecutive slices, "retention" in binary address bits after synchronization, and the grammar-build cost is a separate offline expense. Within each reader contract the stated minima are exact on the reported finite source domain.

Table 4. Separation of acquisition cost, retained coordinate, and reader qualification.

Contract	Prior location information	Worst acquisition	Retained index
Phase 13 sparse, 128 source words	Grid + phase known	11 distant samples	7 bits (j)
Phase 5 sparse, 128 source words	Grid + phase known	15 distant samples	7 bits (j)
Seam reader, 128 source words	Grid + starting phase 62	14 contiguous slices	7 bits (j)
Within-word, 128 source words	Grid + word alignment	62 contiguous slices	7 bits (j)
Sparse, unknown phase	Grid only	55 distant samples	13 bits (j,p)
Contiguous, unknown phase	Grid only; grid competitors	127 contiguous slices	13 bits (j,p)
Contiguous, unknown phase	Grid sources; all ring competitors	256 contiguous slices	13 bits with grid contract

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Contract	Prior location information	Worst acquisition	Retained index
Arbitrary full-ring source	No grid; no phase	Not established for R_{64}	70-bit absolute slice address

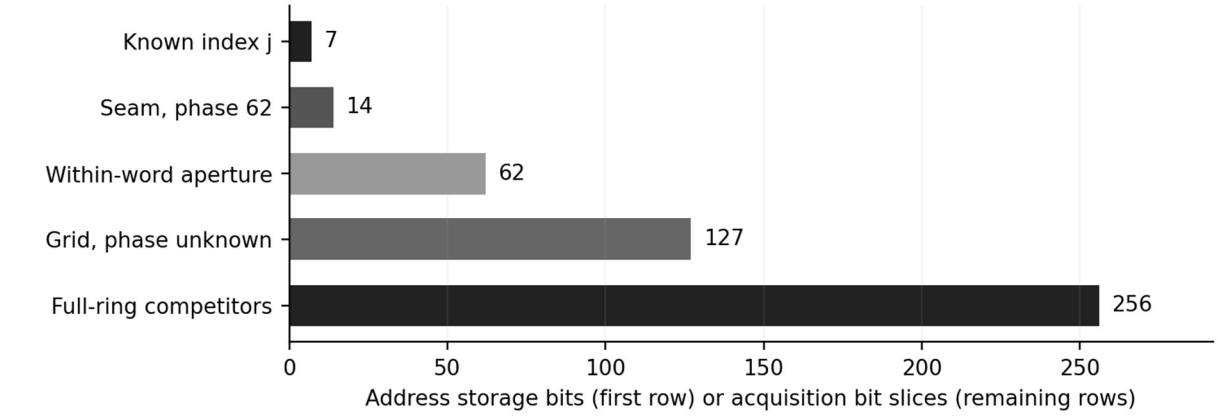


Figure 4. Acquisition-depth comparison across different reader contracts. The seven-bit row is retained address size, not a measurement window. Sparse temporal readers are omitted because their samples have different spacing.

An observation history can be compressed to a route index once the index is determined; that is the continuation-memory claim supported here. It does not erase the time or work needed to synchronize, the storage required for shared rank/unrank code, or the additional winding count required by queries about completed revolutions. The decoder for the 8,192-source grid is a compiled artifact. Building its prefix map costs at least the work of preparing and distinguishing its source signatures; using it later is not equivalent to a uniform polynomial-time construction for arbitrary external data.

14. Limitations, failed extensions, and unresolved links

14.1 A finite sharp threshold is not a general theorem

The selected-source R_{64} result $d(G_{64},\text{full ring})=256$ is exact for the finite source set and the particular native order. It does not establish $d(\text{full ring},\text{full ring})=256$. The width-6 counterexample independently falsifies the earlier universal $4n$ upper bound. Widths 8–17 match $4n$ in their recorded finite census, but the mathematical link needed to prove that pattern for all larger n remains OPEN. The correct next target is a structural bound on the maximum possible alias prefix under the partner-ordered rank/unrank grammar, not another selected numeric fit.

14.2 A reader model is not a physical bandwidth theorem

The main construction is a deterministic discrete mathematical model. The native ring has no inferred material stiffness, spatial embedding, thermal noise law, or causal propagation rate. Its uniform sampling lemmas are arithmetic facts about cyclic groups, and the alias depths reflect known-generator constraints. They neither falsify the sampling theorem for genuinely bandlimited physical waves nor show that the laws of nature necessarily implement this ring order. Assertions that all physical limits arise solely from phase locking are outside the scope of these experiments.

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14.3 Information cannot be regenerated without its grammar

The fact that seven or thirteen address bits regenerate many more output bits depends on the pre-existing full native generator. The generated bits are not independent arbitrary unknown data. Likewise, a reflected phase lane can be computed from one lane only if the partner rule and the palindrome correction are already available; this is conditional reconstruction. Without those conditions, the count of compatible arbitrary binary words remains exponential in the unobserved bits.

14.4 Word histories and exact journeys

Periodicity of the readout destroys information about completed laps. A location index identifies a phase in the current carrier; it does not identify the total number of moves that led there unless that count is retained separately. Distinct ordered recipes can also compose to the same permutation. Therefore the full executable route, its composite operator, its endpoint, and the minimal future-state memory remain different mathematical objects, as demonstrated in the original route-code program [1,2].

14.5 P versus NP and cryptographic inverses

Address resolution for a structured, known ring is not an algorithm for arbitrary SAT, nor an inverse of SHA-256 from its final digest. Constructing and storing the grammar may exploit substantial special structure. Arbitrary inverse queries can lack sufficient data to determine a unique predecessor. The work supports a formal distinction between representation-dependent access and verification but does not decide a classical worst-case complexity separation.

15. Discussion: the continuation ledger as an operational interface

The empirical pattern can be summarized without ontological overreach. The native ring provides a fixed carrier and lawful forward/inverse transformations. A reader defines an observation quotient over that carrier. Longer readouts refine equivalence classes until they become singletons relative to the declared competitor set. At binding, the acquisition history can be replaced by the address class required for subsequent continuation. In the tested grid this retained address contains 7 or 13 bits, even when acquisition requires far larger contiguous windows or many distant samples.

Word boundaries matter because observations that straddle adjacent native words include a relation absent from a within-word aperture. Grid qualification matters because a reader may be comparing its observation only against a constrained set of source addresses or against every address compatible with the full generator. These two facts account for much of the apparent disagreement between previously reported 14-, 62-, 127-, 253- and 256-bit depths. Each number is correct only when paired with its declared source set, comparison set, phase assumption and sampling law.

The theoretical bridge to the older NEXUS result is continuation equivalence: a retained state is not a compressed copy of every historical observation. It is an address class that supports the next legal query. This is compatible with classical automata minimization and symbolic dynamical address coding, while the proposed broader identification of such compilers with physical reality remains a research interpretation requiring independent experimental laws.

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16. Conclusion

A native binary ring with 2^{64} word locations and 2^{70} serialized bit positions can be addressed without enumerating its circumference when its partner ordering and inverse grammar are shared. The exact synchronization depth is not simply its word width: it depends on aperture, phase, stride and the set of admissible competitor origins. For 8,192 fixed source origins on a 128-address R_{64} grid, a phase-unknown contiguous reader requires 127 bits against grid competitors but 256 bits against full-ring competitors, with a 255-bit explicit alias proving sharpness. A sparse unknown-phase reader instead needs 55 widely spaced one-bit samples. All resolve to a 13-bit retained grid coordinate, while known-phase variants resolve to a seven-bit route index. The 3/8 representative boundary and the width-6 exception locate exact structural causes of the longest ambiguous routes. The claims are reproducible within the specified finite domain. The general all-width alias bound and the physical realization of the ring remain open. The sustained result is an operational separation: observation acquires a location; memory preserves only the distinctions needed to continue.

Acknowledgments and statement of scope

All NEXUS constructions and research attribution in this manuscript belong to Dean A. Kulik, QuHarmonics Research Group, ORCID 0009-0003-3128-8828. The calculations are organized from the project's native ring ledgers, standalone programs, execution logs and corrected §178–179 audits. Numerical checks are described with the exact tested domains. This paper does not present the metaphysical Co/C1 interpretation as an independently established physical law.

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Appendix A. Proofs and exact witness definitions

A.1 The partner is an involution

If x is not a palindrome, $J_n(x) \neq x$ and $J_n(J_n(x))=x$, so $P_n^2(x)=x$. If x is a palindrome, $C_n(x)$ is also a palindrome, and $P_n(C_n(x))=C_n(C_n(x))=x$. No word is fixed by P_n because nonpalindromes differ from their reversal and binary complementation changes each bit of a palindrome. Hence the 2^n words form 2^{n-1} unordered partner pairs.

A.2 Candidate coverage for a long unaligned window

Suppose a starting bit phase p is unknown. The number of remaining bits until the next word boundary is $\delta=(n-p) \bmod n$, and δ is at most $n-1$. A window of $W \geq 2n-1$ contains at least n complete bits beginning at that boundary, because $W-\delta \geq n$. Therefore the true boundary alignment is among $\delta=0,\ldots,n-1$. Ranking the extracted native word returns its actual word address, so the true absolute bit start is a candidate of Algorithm 1. The algorithm may contain aliases but cannot omit the true source under the stated shared-grammar assumptions.

A.3 The 3/8 representative count for even widths

Partition leading-zero words into last-bit-one (2^{n-2} words) and last-bit-zero (2^{n-2} words). Every leading-zero, last-bit-one word is lexicographically less than its reverse, giving 2^{n-2} canonical representatives. The 0…0 class consists of reversal two-cycles and palindromes. It contains $Q=2^{n-2}$ words and $P=2^{n/2-1}$ palindromes. Its canonical count is $(Q-P)/2+P=(Q+P)/2=2^{n-3}+2^{n/2-2}$. Adding 2^{n-2} proves equation (14). The complement-paired palindrome rule ensures that the leading-zero palindrome is the canonical member in each such pair.

A.4 Width-6 certificate

Source t=84 begins at word address a=14 with phase 0; competitor u=314 begins at a=52 with phase 2. Their 24-bit observations are identical and given verbatim in Figure 1. Their 25th bits disagree. The companion native-window execution log certifies injectivity of all 384 observed 25-bit windows. Thus an observation of length 24 is insufficient but 25 is sufficient, proving $w^*(6)=25$.

A.5 Why the last R_{64} selected-source alias gives an exact threshold

By equation (8), an alias sharing 255 initial bits establishes the lower bound $W \geq 256$. The full §178 8,192×64 alignment audit found no alternative full-ring candidate agreeing with any tested source for 256 bits. Because every sufficiently long candidate reading must contain a complete native word and hence appear in one of the 64 rank-based alignments, the absence of surviving candidates establishes sufficiency. Both statements apply to the selected source set G_{64} only; the all-source full-ring problem remains independent.

A.6 Phase-memory and winding

Given a known source index j on the 128-position grid, every subsequent sparse sample index is (j+k) mod 128. If a query asks how many complete 128-sample revolutions elapsed, the return count $q=\lfloor (j+k)/128 \rfloor$ or an external equivalent must be retained. Without that counter, routes differing by integral full laps share the same current phase and future periodic readout. This is a statement about the query's required memory, not a failure of the cycle to be reversible.

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Appendix B. Reproducibility manifest and correction ledger

B.1 Executable evidence in the supplied archive

Every program below is source material authored and executed within the NEXUS research record. The full paper does not silently collapse a source transcript, code assertion, and independent theorem into one evidentiary category.

Table 5. Reproducibility manifest for the NEXUS §§170–179 executable record.

Stage	Verification file	Primary evidence
§170	NEXUS_170_native_phase_router.py	Unknown-boundary rank/unrank; width-17 depth and small-ring returns
§171	NEXUS_171_recipe_noun_substitution.py	Payload-independent reversible recipe and cycle classes
§172	NEXUS_172_full_circumference_sampling.py	Full circumference, stride and sampling phase law
§173	NEXUS_173_reflected_sampling_lanes.py	Partner reflection and palindrome correction
§174	NEXUS_174_native_transverse_compiler_standalone.py	Native rank/increment/unrank successor and dependency matrix
§175	NEXUS_175_memory_shape.py	Known-phase sparse reader, 15 samples and seven-bit index
§176	NEXUS_176_seam_memory_compiler.py	Fourteen-bit seam reader, 62-bit within-word aperture
§177	NEXUS_177_self_locating_reader.py	Unknown-phase 8,192-source sparse/contiguous readers
§178	NEXUS_178_full_ring_decompiler.py	Unfiltered 8,192-source census and 255-bit alias
§179	NEXUS_179_native_boundary_audit.py	Width-6 counterexample and triple-owner witnesses

B.2 Corrections that must remain attached to citation

Table 6. Correction ledger retained with the source archive.

Correction locus	Corrected statement
Source grid versus all-source ring	§178 checked 8,192 source origins against all ring competitors; it did not check all 2^{70} sources.
Width-6 exception	$w^*(6)=25$ disproves an unqualified all-width $W \le 4n$ claim.
Representative count exponents	The corrected $n=64$ count is $3\cdot 2^{61}+2^{30}$, not $3\cdot 2^{62}+2^{31}$.
Phase-5 14-bit prefix statistic	126 distinct signatures means 124 uniquely identified starting positions, not 126.
Nyquist terminology	The 127/128-bit synchronization horizons are not Nyquist bandwidth equalities.
Two clocks	τ_n and the full native address successor T_n are not the same operator.
Evidence state	The initial §178 seal is preserved with its wrong exponents and the subsequent explicit correction; §179 is retrospective.

B.3 Data-access and publication notes

The §178 archive supplies the 2,451-row alias census, native decompiler, exact outputs, and figure; §179 supplies the width-6 and triple-owner witnesses. Replication requires identical ring ordering, declared comparison domains, and the original correction logs. Publication identifiers may be added after archival deposit.

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